

SJAA Ephemeris Bulletin



Dec 2025

Reflector

As 2025 draws to a close, let's take a quick look back at some of the year's astronomical highlights!

As we noted last month, October gave us some great views of Comets C/2025 A6 Lemmon and C/2025 R2 Swan. We also were visited by the third known interstellar object, 3I/ATLAS. Orbital space launches topped out at 292, with only 11 failures. In January, the Blue Ghost Mission 1 by Firefly Aerospace successfully landed on the Moon, carrying NASA-sponsored experiments as well as commercial payloads. In February, ESA's Solar Orbiter broke free from the orbital plane of the Solar System to photograph the Sun's south pole: a first from that vantage point.

Of course, SJAA had its own share of highlights. For me, our Show and Tell event was a big one. Members showed up in force to share their experiences in astrophotography, DIY optics and telescopes, and even some song and dance! The Yosemite Star Party in July was an unforgettable and rewarding public star party. Multiple Swap Meets throughout the year brought new and old friends together to bond over equipment.

And there's more to come! SJAA has a full calendar for 2026 with all of your favorite programs returning and then some. I can't wait to share in the fun with you all. Until then, keep looking up!

Carl Svensson
Vice-President
Ephemeris Editor

Target of the Month: Pleiades

By Joe Fragola The Pleiades, also known as Messier 45 or the Seven Sisters, is a bright open star cluster easily visible to the naked eye. This small, sparkling cluster of stars is located within the constellation Taurus, the Bull. In early 1610 Galileo Galilei was the first astronomer to observe the Pleiades through a telescope. Charles Messier cataloged the Pleiades as Messier 45 on March 4, 1769. It is among the nearest star clusters to Earth and is the nearest Messier object to Earth. The cluster rises in the East during the fall and early winter evenings and is best viewed between October and April when it's high in the night sky.

In a dark, clear sky, most people can see six or seven stars in the cluster, though some may spot even more. The Pleiades look like a tiny dipper or a small cloud of sparkling lights. Using binoculars, dozens of stars come into view, and you may see a blue haze surrounding the brightest members. A small telescope reveals even more stars, and you might catch hints of nebulosity - the faint glow of dust and gas reflecting starlight.

The Pleiades have inspired legends in many cultures. In ancient Greece, they were the seven daughters of the Titan Atlas and the sea-nymph Pleione, pursued across the heavens by Orion the Hunter. In Japan, the cluster is called "Subaru," meaning "to gather together". Native American tribes, such as the Lakota and the Cherokee, also have stories about the Pleiades - often portraying them as seven children who found safety among the stars.

To find the Pleiades, look for the constellation Orion, which is easy to spot thanks to his "belt" a group of three stars in a straight line. Trace a line upward from Orion's belt, past the bright reddish star Aldebaran (the eye of Taurus), to a small, fuzzy cluster of stars. This is the Pleiades.

Magnitude	1.6
Size	43 ly
Object Type	Open Star Cluster
Constellation	Taurus
Distance	444 ly
Age	150 million years



Pleiades image captured on 11-21-2025 at Houge Park by Joe Fragola; collection time of 20 minutes (121 10-second subframes) using a ZWO Seestar S50. No post processing was done.

Announcements

Upgraded Email Lists The Observers and Astroimaging email lists are getting new names! They will continue to thrive under slightly new addresses: observers@sjaa.net and astroimaging@sjaa.net. This move allows us to give you more control over finding and joining our email lists. Head over to your profile on membership.sjaa.net, and sign yourself up for these groups if you aren't already in them. Don't forget to update your contacts with the new addresses!

We Always Need More Volunteers There are many volunteer opportunities in SJAA! While we love having

seasoned experts in our volunteer ranks, we happily welcome enthusiastic beginners, too. Opportunities range from operating telescopes, to authoring content for publication, to helping with event logistics. Please reach out to us at volunteer@sjaa.net to get started!

Member Survey Results Back in the spring and early summer, we conducted a survey of our members to better understand what folks like about SJAA, and what can be improved. We learned a lot. Check out Wolf's article in the December issue of the Ephemeris for more info.



SJAA Dec 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1  87%	2 Comet C/2025 T1 Perihelion Astronomy Beginners' Forum 7:00PM  94%	3 Moon 0.8° N of Pleiades New Member Meet and Greet 6:30PM  98%	4 Full Moon Astro 101: All About Binoculars 7:00PM  100%	5  99%	6 Board Meeting 6:00PM Guest Speaker 7:30PM  95%
7 Gerard Kuiper's birthday Jupiter 4° S of Moon  89%	8 Moon 1.4° N of Beehive star cluster  81%	9 Regulus 0.7° S of Moon  72%	10  62%	11 Annie Jump Cannon's birthday Last Quarter Moon  51%	12  41%	13 Starry Nights Star Party 6:00PM  32%
14 Tycho Brahe's birthday Spica 1.4° N of Moon  23%	15  15%	16 Edward Emerson Barnard's birthday Imaging SIG 7:30PM  9%	17  5%	18 Antares 0.4° N of Moon  2%	19 New Moon Astro 101 6:00PM  0%	20 Member Observing Night 4:30PM  1%
21 Winter Solstice  3%	22 Pluto 0.6° S of Moon  7%	23  13%	24  21%	25 Isaac Newton's birthday  30%	26 Saturn 4° S of Moon  41%	27 Johannes Kepler's birthday Neptune 3° S of Moon  52% First Quarter Moon
28 Arthur Eddington's birthday  64%	29 Maarten Schmidt's birthday  74%	30  84%	31 Moon 0.9° N of Pleiades  91%			
			Times and dates are subject to change. Please check sjaa.net and www.meetup.com/sj-astronomy for the latest updates.			
						 Meteor Shower  Dark Sky Window